

SIMATS ENGINEERING ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

Duration : 45 Minutes
Total Marks:20

QUESTIONS: 2

Annexure A

Descriptive Written Test

Code of Conduct:

I S. MD Imthiah (192521366) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

S. Imthiah
Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different

(i) Uniform Cost Search (UCS)

⇒ It expands the node with low path cost

Working:

1. Start from the initial position
2. Calculate path cost.
3. Expand the lowest-cost path.
4. Continue until the exit is reached.

Advantage:

- * Finds the optimal (low-cost) path.
- * Works well when path costs differ.

Disadvantages:

- * High memory usage
- * Slower for very large maps.

Complexity:

- * Time : $O(b^{1 + (C/\epsilon)})$
- * Space : $O(b^{1 + C^*/(\epsilon)})$.

b → branching factor.

C → optimal path cost

ϵ ⇒ minimum step cost.

Iterative Deepening Search (IDS)

Working:

- * Search depth = 0
- * Then depth = 1
- * Then depth = 2
- * Continue until the exit is found

Advantage:

- * Low memory usage
- * Complete algorithm

Disadvantage:

- * Repeats previous search.
- * Slower due to repeated expansions.

Complexity:

* Time : $O(b^d)$

* Space : $O(bd)$

where :

b = branching factor

d = Solution depth.

A robot is placed in an unknown environment and must find a path to exit.

Relation with intelligent agent

1) Simple Reflex Agent:

- It responds only to the given input.
- It does not maintain any state for complex system.

2) Goal based Agent:

- It focuses only on the path to reach the specific goal.

3) Model based

- It stores the previous state function and updates the knowledge on new input.

4) Utility Based

- It is based on least cost path.

It takes decision path which has lowest path cost.

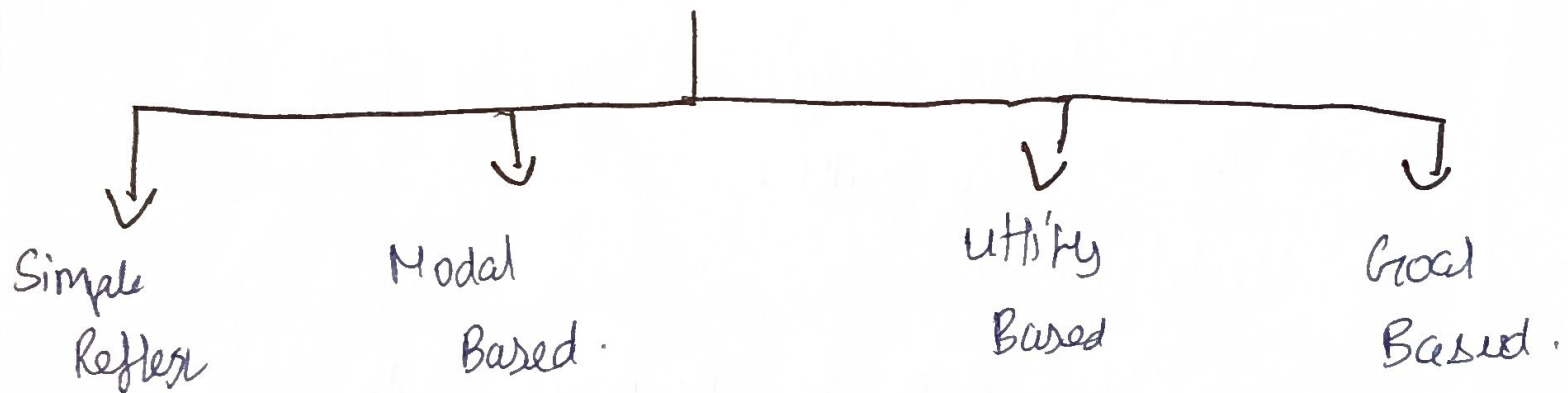
Combination:

• I suggest Uniform Cost search with goal based agent because UCS finds the minimal cost path and goal Agent choose the best path to reach the goal.

1. Smart home Automations System.

⇒ Smart Home Automation System is an application AI. that controls the lighting, temperature and security based on user behaviour and environmental condition. Intelligent Agent helps the system to take decision.

Types of Agent.



* Simple Reflex Agent.

* Make decisions based only on the current input ~~and~~ it perform the predefined action.

Ex:

Motion detected → Turn On light.

* Model Based Agent.

* It stores the previous state in internal memory and updates its knowledge by using new input.

Ex: The system remembers the previous state at what time he came inside the room.

Goal Based Agent-

⇒ Takes action to achieve the specific goals.

Ex:

Goal = Save energy Turn off room light.

⇒ Utility Based Agent

+ It selects the best and low cost path which will give profit or less.

Hybrid Approach

⇒ It is a combination of Model based, Goal based, and Utility Based.

⇒ It stores the previous data selects the low cost path to achieve the specific goal.

That is.

Ex:

When the person enters the room it knows that the person will enter the room by 10 PM

at that time it closes the door, turn on the

AC & based on the ~~electric~~ peak hours of electricity

it adjust the temperature and less power consumption.

types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 Proficient	Level 2 - Developing	Level 1 Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand